

Experiment No. 8
Apply FP based estimation technique for a case study
Date of Performance:
Date of Submission:

Aim: To apply FP based estimation technique for a case study.

Objective: To analyse and apply FP based estimation technique for Currency Detector for the Visually Impaired.

Theory:

Function Point (FP) is an element of software development which helps to approximate the cost of development early in the process. It may measure functionality from user's point of view.

Objectives of FPA:

- The objective of FPA is to measure the functionality that the user requests and receives.
- The objective of FPA is to measure software development and maintenance independently of the technology used for implementation.
- It should be simple enough to minimize the overhead of the measurement process.
- It should be a consistent measure among various projects and organizations.

Types of FPA:

□ **Transactional Functional Type –**

□ **External Input (EI):** EI processes data or control information that comes from outside the application's boundary. The EI is an elementary process.

□ **External Output (EO):** EO is an elementary process that generates data or control information sent outside the application's boundary.

□ **External Inquiries (EQ):** EQ is an elementary process made up of an input-output combination that results in data retrieval.

□ **Data Functional Type –**

□ **Internal Logical File (ILF):** A user identifiable group of logically related data or control information maintained within the boundary of the application.

□ **External Interface File (EIF):** A group of users recognizable logically related data allusion to the software but maintained within the boundary of another software.

STEPS:

□ **Step-1:**

$F = 14 * \text{scale}$

Scale varies from 0 to 5 according to character of Complexity Adjustment Factor (CAF). Below table shows scale

- 0 - No Influence
- 1 - Incidental
- 2 - Moderate
- 3 - Average
- 4 - Significant
- 5 - Essential

□ **Step-2:** Calculate Complexity Adjustment Factor (CAF).

$$CAF = 0.65 + (0.01 * F)$$

□ **Step-3:** Calculate Unadjusted Function Point (UFP).

Multiply each individual function point to corresponding values in TABLE.

□ **Step-4:** Calculate Function Point.

$$FP = UFP * CAF$$

Example:

Given the following values, compute function point when all complexity adjustment factor (CAF) and weighting factors are average.

User Input = 50

User Output = 40

User Inquiries = 35

User Files = 6

External Interface = 4

Explanation:

□ **Step-1:** As complexity adjustment factor is average (given in question), hence,

□ scale = 3.

$$F = 14 * 3 = 42$$

□ **Step-2:**

$$CAF = 0.65 + (0.01 * 42) = 1.07$$

□ **Step-3:** As weighting factors are also average (given in question) hence we will multiply each individual function point to corresponding values in TABLE.

$$UFP = (50*4) + (40*5) + (35*4) + (6*10) + (4*7) = 628$$

□ **Step-4:**

$$\text{Function Point} = 628 * 1.07 = \mathbf{671.96}$$

Code:

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in); // Create a Scanner object
        System.out.println("=====\n::Functional Point Cost
Estimation::\n=====\n");
        System.out.println("Enter the following details for your project:");
        System.out.print("Number of user input:");
        int ip = myObj.nextInt(); // Read user input
        System.out.print("Number of user output:");
        int op = myObj.nextInt();
        System.out.print("Number of user enquires:");
        int eq = myObj.nextInt();
        System.out.print("Number of user files:");
```

```

int fl = myObj.nextInt();
System.out.print("Number of user external interfaces:");
int ifs = myObj.nextInt();
System.out.print("\nchoose the complexity (1.SIMPLE 2.AVERAGE 3.COMPLEX):");
int compx = myObj.nextInt();
int count_total=0;
int[] Simple = new int[]{3,4,3,7,5};
int[] avg = new int[]{4,5,4,10,7};
int[] complex = new int[]{6,7,6,15,10};

System.out.print("\nComplexity Factor:");
double CF = myObj.nextDouble();

if (compx==1){
    count_total=ip*Simple[0]+op*Simple[1]+eq*Simple[2]+fl*Simple[3]+ifs*Simple[4];
}
if (compx==2){
    count_total=ip*avg[0]+op*avg[1]+eq*avg[2]+fl*avg[3]+ifs*avg[4];
}
if (compx==3){
    count_total=ip*complex[0]+op*complex[1]+eq*complex[2]+fl*complex[3]+ifs*complex[4];
}

double FP=count_total*CF;
System.out.print("\n\nFunctional Point= "+FP+"");

}
}

```

Output:

```

=====
Functional Point Cost Estimation::
=====
Enter the following details for your project:
Number of user input:8
Number of user output:8
Number of user enquires:2
Number of user files:2
Number of user external interfaces:2
choose the complexity (1.SIMPLE 2.AVERAGE 3.COMPLEX):2
Complexity Factor:1.40
Functional Point= 159.6

```

Conclusion: We have successfully applied FP estimation technique for Currency Detector for the Visually Impaired.